PRODUCT USER MANUAL

For Sea Level Altimeter products

SEALEVEL_GLO_PHY_CLIMATE_L4_MY_008_057

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/07/27	all	Creation	M Pujol	
1.1	2021/01/15	**Erreur ! Source du renvoi introuvable.**	Update of C3S web links	M Pujol	M Pujol
2.0	2021/09/10		Update with DT-2021 standards	G Taburet	M Pujol
2.1	2021/12/16	**Erreur ! Source du renvoi introuvable.**	Change for MY automatic extensions	M-I Pujol	M-I Pujol
3.0	2023/08/03		Change for MY interim production Use new template	M-I Pujol	M-I Pujol
4.0	2024/07/17		Update for Nov2024 version (DT-2024 standards)	M-I Pujol	M-I Pujol

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GLOSSARY AND ABBREVIATIONS

ADT	Absolute Dynamic Topography
CF	Climate Forecast (convention for NetCDF)
CLS	Collecte Localisation Satellite
CNES	Centre National d'Etude Spatiale (French Space Agency)
CPU	Central Processing Unit
DAC	Dynamic Atmospheric Correction
DGF	Direct Get File
DT	Delayed Time
DUACS	Data Unification Altimeter Combination System
EUMETSAT	European operational satellite agency
GDR	Geophysical Data Record
GLO	Global Ocean
HF	High Frequency
HR	High Resolution
IGDR	Intermediate Geophysical Data Record
L2	Niveau 2
L2P	Niveau 2P
L3	Niveau 3
L4	Niveau 4
LRM	Low Resolution Mode
LWE	Long Wavelength Error
MADT	Map of ADT
MDT	Mean Dynamic Topography
MP	Mean Profile
MSLA	Map of Sea Level Anomaly
MSS	Mean Sea Surface
MY	Multi-Year
NetCDF	Network Common Data Form
NFS	Network File System
NRT	Near Real Time
NTC	None Time Critical (same as GDR)
OGDR	Operational Geophysical Data Record
QUID	Quality Information Document
REP	Reprocessing (MY)
RT	Real Time
SALP	Service d'Altimetrie et Localisation Precise (altimetry and precise positioning service)
SAR	Synthetic Aperture Radar
SL	Sea Level
SLA	Sea Level Anomaly
SL-TAC	Sea Level Thematic Assembly Centre
SQO	Scientific Quality Overwiev
SSALTO	Segment Sol multimissions d'ALTimétrie, d'Orbitographie et de localisation précise
SSH	Sea Surface Height
STC	Short Time Critical (same as IGDR)

TAC	Thematic Assembly Center
WMS	Web Map Service

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I. INTRODUCTION

I.1 Summary

The Sea Level TAC (Thematic Assembly Centre) is one of the eight TAC of the Copernicus Marine Service. Part of the production also concerns the Copernicus Climate Change Service (C3S) sea level production, also available in the C3S catalogue. The aim of this document is to describe this product.

The data produced in the frame of this service are generated by the processing system including data from all altimeter Copernicus missions (Sentinel-6A, Sentinel-3A) and other collaborative or opportunity missions (e.g. :Jason-3, Saral[-DP]/AltiKa, Cryosat-2, OSTM/Jason-2, Jason-1, Topex/Poseidon, Envisat, ERS-1/2).

The main products lines consist in **Level-4 (L4; gridded) altimeter products** produced in **Near Delayed Time/ Multi-Year (DT/MY)**. These datasets benefit from the highest quality altimeter measurements and geophysical corrections, and are produced with a unique system (no changes in time in the set up, extended in time using the same system) to minimize the risk of quality loss or spurious signals appearing in time.

The data provided to users have a **global coverage.**

The user can check the version of the product used with the global attributes given in netCDF files:

- product_version : gives the information of the main Copernicus Marine release for which the last change has been implemented (e.g. "vDec2021", “vNov2024”)
- software_version : gives the precise software version used for the production, and information of the reference altimeter standards and processing used (e.g. : "7.0_DUACS_DT2021_baseline")

II. DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Product Lines	SEALEVEL_GLO_PHY_CLIMATE_L4_MY_008_057
Geographical coverage	GLO : Global Ocean
Variables	The different variables included in this product are listed in section **Erreur ! Source du renvoi introuvable.**.
Product Type	MY : Multi Year
Available time series	MY : 1993/01/01 ; -5 to 9M before present.
Temporal resolution	Daily
Target delivery time	MY : ~4 times a year
Delivery mechanism	Marine Information System
Horizontal resolution	GLO : 0.25°x0.25°
Number of vertical levels	1
Format	Netcdf CF1.6

Table 1: C3S L4 products

II.2 Details of datasets

SEALEVEL_GLO_PHY_CLIMATE_L4_MY_008_057
Dataset: **c3s_obs-sl_glo_phy-ssh_my_twosat-l4-duacs-0.25deg_P1D**
Physical variables name in the NetCDF file and Unit: Long_name & Standard_name
sla [m] Sea level anomaly sea_surface_height_above_sea_level
err_sla [m] Formal mapping error sea_surface_height_above_sea_level standard_error
ugosa [m/s]

Geostrophic velocity anomalies: zonal component surface_geostrophic_eastward_sea_water_velocity_assuming_sea_level_for_geoid
err_ugosa [m/s] Formal mapping error on zonal geostrophic velocity anomalies surface_geostrophic_eastward_sea_water_velocity_assuming_sea_level_for_geoid standard_error
vgosa [m/s] Geostrophic velocity anomalies: meridian component surface_geostrophic_northward_sea_water_velocity_assuming_sea_level_for_geoid
err_vgosa [m/s] Formal mapping error on meridional geostrophic velocity anomalies surface_geostrophic_northward_sea_water_velocity_assuming_sea_level_for_geoid standard_error
adt [m] Absolute dynamic topography sea_surface_height_above_geoid
ugos [m/s] Absolute geostrophic velocity: zonal component surface_geostrophic_eastward_sea_water_velocity
vgos [m/s] Absolute geostrophic velocity: meridian component surface_geostrophic_northward_sea_water_velocity
Coordinate variables name in the NetCDF file and Unit: Long_name & Standard_name
time [days since 1950-01-01 00:00:00] Time of measurement time
longitude [degrees_east] Longitude of measurement longitude
latitude [degrees_north] Latitude of measurement latitude
lon_bnds [degrees_east] - -
Lat_bnds[degrees_north] - -
Other useful variables name in the NetCDF file and Unit: Long_name & Standard_name
flag_ice [0,1] Ice Flag for a 15% criterion of ice concentration status_flag
tpa_correction [m] TOPEX-A instrumental drift correction derived from altimetry and tide gauges global comparisons sea_surface_height_above_sea_level
Dataset: **c3s_obs-sl_glo_phy-ssh_my_twosat-l4-duacs-0.25deg_P1M-m**
Physical variables name in the NetCDF file and Unit: Long_name & Standard_name
sla [m]

Averaged Sea Level Anomalies sea_surface_height_above_sea_level
eke [cm²/s²] Averaged Eddy Kinetic Energy specific_kinetic_energy_of_sea_water
Coordinate variables name in the NetCDF file and Unit: Long_name & Standard_name
time [days since 1950-01-01 00:00:00] Time of measurement time
longitude [degrees_east] Longitude of measurement longitude
latitude [degrees_north] Latitude of measurement latitude
lon_bnds [degrees_east] - -
Lat_bnds[degrees_north] - -

Table 2 : List of the datasets and variable names and unit for the SEALEVEL_GLO_PHY_CLIMATE_L4_MY_008_product

II.3 Additional Information on parameters

sla	Sea Level Anomaly (see Annex)
err_sla err_ugosa err_vgosa	Formal mapping error (see QUID document) associated to sla, ugosa and vgosa variables
ugosa & vgosa ugos & vgos	Anomalies of the geostrophic current and absolute geostrophic current components (see QUID document) in the zonal and meridian directions
mdt	Mean Dynamic Topography (see Annex)
adt	Absolute Dynamic Topography (see Annex)
tpa_correction	TOPEX-A instrumental drift correction derived from altimetry and tide gauges global comparisons. This variable can be added to the SLA to correct for the observed instrumental drift during the lifetime of the TOPEX-A mission (the correction is null after this period). This is a global mean correction to be added a posteriori (and not before) on the global mean sea level estimate derived from the sea level variable. It can be

	applied at regional or local scale as a best estimate (better than no correction, since the regional variation of the instrumental drift is unknown) Global_MSLTPA_drift_corrected(d) = Global_MSLdeduced from SLA from product(d) + tpa_correction(d). Where d is the date of the day considered
eke	Eddy kinetic energy deduced from the anomalies of the geostrophic current: eke = 0.5*(ugosa² + vgosa²)

II.4 Production System Description

The SEALEVEL altimeter products are processed with the CNES/CLS DUACS (Data Unification and Altimeter Combination System) system.

The DUACS processing steps are described in the QUID document.

The altimeter measurements used as input of the DUACS MY system consist in a majority of Level 2p products from different missions, that are available in Delayed Time or Non-Time Critical product (GDR or NTC).

Details of the different L2p altimeter products sources and delay of availability are given in the QUID document.

II.5 Grid

All L4 products are delivered onto cartesian grids. All the variables are defined at the center of the output grid cell.

II.6 Vertical Levels

All the SEALEVEL products are defined on the surface layer.

II.7 Processing information

The products/datasets are extended in time four times a year, so that these datasets cover from 1993 up to five to nine months before present. These datasets benefit from the highest quality altimeter measurements and geophysical corrections, and are produced with a unique system (no changes in time in the set up, extended in time using the same system) to minimize the risk of quality loss or spurious signals appearing in time.

A full reprocessing of the Multi-Year series is usually delivery each 3 to 4 years.

III. FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV. FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

III.1 Nomenclature of files when downloaded through the Copernicus Marine Web Portal Subset Service

Files nomenclature when downloaded through the Web Portal Subset is based on product dataset name and a numerical reference related to the request date on the MIS.

The scheme is: **datasetname_nnnnnn.nc**

where :

- **datasetname** is a character string within one of the datasets listed in section **Erreur ! Source du renvoi introuvable.** and **Erreur ! Source du renvoi introuvable.** :
- **nnnnnn**: 6 digits integer corresponding to the current time (download time) in milliseconds since January 1, 1970 midnight UTC.
- **nc**: standard NetCDF filename extension.

Example:

c3s_obs-sl_glo_phy-ssh_my_twosat-l4-duacs-0.25deg_P1D_1644250104522.nc

IV.2 Nomenclature of original files

The nomenclature used is:

<delay>_<zone>_twosat_phy_l4_<datemap>_<dateprod>.<format>

where the fields in "<>" are described below:

delay	dt	delayed time products
zone	global	global geographic coverage product
date	YYYYMMDD	date of the map
dateprod	YYYYMMDD	production date of the map
format	.nc	NetCdf CF1.6

IV.3 File size

The following table give an estimation of the typical size of one the daily file for L3 and L4 products, when downloaded through the Marine Data Store.

Area	**Product Level**	**NetCDF file size**
Global	L4	9.2 Mo

IV.4 Structure of files

Examples of the header of output NetCDF files are inserted in annex, for each dataset.

V. REFERENCES

QUID : the associated QUID document is CMEMS-SL-QUID-008-057

C3S PUGS document : available at https://cds.climate.copernicus.eu/cdsapp#!/dataset/satellite-sea-level-global?tab=doc

VI. ANNEX

An example output NetCDF file header for the cmems_obs-sl_glo_phy-ssh_my_allsat-l4-duacs-0.25deg_P1D dataset is given bellow.

```
netcdf dt_global_twosat_phy_l4_20220723_vDT2021{
dimensions:
    time = 1 ;
    latitude = 720 ;
    longitude = 1440 ;
    nv = 2 ;
variables:
    int crs ;
        crs:comment = "This is a container variable that describes the grid_mapping used by the data in this file. This variable does not contain anydata; only information about the geographic coordinate system." ;
        crs:grid_mapping_name = "latitude_longitude" ;
        crs:inverse_flattening = 298.257 ;
        crs:semi_major_axis = 6378136.3 ;
    float time(time) ;
        time:axis = "T" ;
        time:calendar = "gregorian" ;
        time:long_name = "Time" ;
        time:standard_name = "time" ;
        time:units = "days since 1950-01-01 00:00:00" ;
    float latitude(latitude) ;
        latitude:axis = "Y" ;
        latitude:bounds = "lat_bnds" ;
        latitude:long_name = "Latitude" ;
        latitude:standard_name = "latitude" ;
        latitude:units = "degrees_north" ;
        latitude:valid_max = 89.875 ;
        latitude:valid_min = -89.875 ;
    float lat_bnds(latitude, nv) ;
        lat_bnds:comment = "latitude values at the north and south bounds of each pixel." ;
        lat_bnds:units = "degrees_north" ;
    float longitude(longitude) ;
        longitude:axis = "X" ;
        longitude:bounds = "lon_bnds" ;
        longitude:long_name = "Longitude" ;
        longitude:standard_name = "longitude" ;
        longitude:units = "degrees_east" ;
        longitude:valid_max = 179.875 ;
        longitude:valid_min = -179.875 ;
    float lon_bnds(longitude, nv) ;
        lon_bnds:comment = "longitude values at the west and east bounds of each pixel." ;
        lon_bnds:units = "degrees_east" ;
    int nv(nv) ;
        nv:comment = "Vertex" ;
        nv:long_name = "Number of cell vertices" ;
        nv:units = "1" ;
    int sla(time, latitude, longitude) ;
        sla:_FillValue = -2147483647 ;
        sla:ancillary_variables = "err_sla" ;
        sla:comment = "The sea level anomaly is the sea surface height above mean sea surface; it is referenced to the [1993, 2012] period; see the poduct user manual for details" ;
        sla:coordinates = "longitude latitude" ;
        sla:grid_mapping = "crs" ;
        sla:long_name = "Sea level anomaly" ;
        sla:scale_factor = 0.0001 ;
        sla:standard_name = "sea_surface_height_above_sea_level" ;
        sla:units = "m" ;
    int err_sla(time, latitude, longitude) ;
        err_sla:_FillValue = -2147483647 ;
```

```
        err_sla:comment = "The formal mapping error represents a purely theoretical mapping error. It mainly traduces errors induced by the constellaion sampling capability and consistency with the spatial/temporal scales considered, as described in Le Traon et al (1998) or Ducet et al (2000)" ;
        err_sla:coordinates = "longitude latitude" ;
        err_sla:grid_mapping = "crs" ;
        err_sla:long_name = "Formal mapping error" ;
        err_sla:scale_factor = 0.0001 ;
        err_sla:standard_name = "sea_surface_height_above_sea_level standard_error" ;
        err_sla:units = "m" ;
    int ugosa(time, latitude, longitude) ;
        ugosa:_FillValue = -2147483647 ;
        ugosa:ancillary_variables = "err_ugosa" ;
        ugosa:comment = "The geostrophic velocity anomalies are referenced to the [1993, 2012] period" ;
        ugosa:coordinates = "longitude latitude" ;
        ugosa:grid_mapping = "crs" ;
        ugosa:long_name = "Geostrophic velocity anomalies: zonal component" ;
        ugosa:scale_factor = 0.0001 ;
        ugosa:standard_name = "surface_geostrophic_eastward_sea_water_velocity_assuming_sea_level_for_geoid" ;
        ugosa:units = "m/s" ;
    int err_ugosa(time, latitude, longitude) ;
        err_ugosa:_FillValue = -2147483647 ;
        err_ugosa:comment = "The formal mapping error represents a purely theoretical mapping error. It mainly traduces errors induced by the constelation sampling capability and consistency with the spatial/temporal scales considered, as described in Le Traon et al (1998) or Ducet et al (2000)" ;
        err_ugosa:coordinates = "longitude latitude" ;
        err_ugosa:grid_mapping = "crs" ;
        err_ugosa:long_name = "Formal mapping error on zonal geostrophic velocity anomalies" ;
        err_ugosa:scale_factor = 0.0001 ;
        err_ugosa:standard_name = "surface_geostrophic_eastward_sea_water_velocity_assuming_sea_level_for_geoid standard_error" ;
        err_ugosa:units = "m/s" ;
    int vgosa(time, latitude, longitude) ;
        vgosa:_FillValue = -2147483647 ;
        vgosa:ancillary_variables = "err_vgosa" ;
        vgosa:comment = "The geostrophic velocity anomalies are referenced to the [1993, 2012] period" ;
        vgosa:coordinates = "longitude latitude" ;
        vgosa:grid_mapping = "crs" ;
        vgosa:long_name = "Geostrophic velocity anomalies: meridian component" ;
        vgosa:scale_factor = 0.0001 ;
        vgosa:standard_name = "surface_geostrophic_northward_sea_water_velocity_assuming_sea_level_for_geoid" ;
        vgosa:units = "m/s" ;
    int err_vgosa(time, latitude, longitude) ;
        err_vgosa:_FillValue = -2147483647 ;
        err_vgosa:comment = "The formal mapping error represents a purely theoretical mapping error. It mainly traduces errors induced by the constelation sampling capability and consistency with the spatial/temporal scales considered, as described in Le Traon et al (1998) or Ducet et al (2000)" ;
        err_vgosa:coordinates = "longitude latitude" ;
        err_vgosa:grid_mapping = "crs" ;
        err_vgosa:long_name = "Formal mapping error on meridional geostrophic velocity anomalies" ;
        err_vgosa:scale_factor = 0.0001 ;
        err_vgosa:standard_name = "surface_geostrophic_northward_sea_water_velocity_assuming_sea_level_for_geoid standard_error" ;
        err_vgosa:units = "m/s" ;
    int adt(time, latitude, longitude) ;
        adt:_FillValue = -2147483647 ;
        adt:comment = "The absolute dynamic topography is the sea surface height above geoid; the adt is obtained as follows: adt=sla+mdt where mdt i the mean dynamic topography; see the product user manual for details" ;
        adt:coordinates = "longitude latitude" ;
        adt:grid_mapping = "crs" ;
        adt:long_name = "Absolute dynamic topography" ;
        adt:scale_factor = 0.0001 ;
        adt:standard_name = "sea_surface_height_above_geoid" ;
        adt:units = "m" ;
    int ugos(time, latitude, longitude) ;
        ugos:_FillValue = -2147483647 ;
        ugos:coordinates = "longitude latitude" ;
        ugos:grid_mapping = "crs" ;
        ugos:long_name = "Absolute geostrophic velocity: zonal component" ;
        ugos:scale_factor = 0.0001 ;
        ugos:standard_name = "surface_geostrophic_eastward_sea_water_velocity" ;
```

```
        ugos:units = "m/s" ;
    int vgos(time, latitude, longitude) ;
        vgos:_FillValue = -2147483647 ;
        vgos:coordinates = "longitude latitude" ;
        vgos:grid_mapping = "crs" ;
        vgos:long_name = "Absolute geostrophic velocity: meridian component" ;
        vgos:scale_factor = 0.0001 ;
        vgos:standard_name = "surface_geostrophic_northward_sea_water_velocity" ;
        vgos:units = "m/s" ;
    int tpa_correction(time) ;
        tpa_correction:_FillValue = -2147483647 ;
        tpa_correction:comment = "This variable can be added to the gridded SLA to correct for the observed instrumental drift during the lifetime ofthe TOPEX-A mission (the correction is null after this period). This is a global correction to be added a posteriori (and not before) on the global mean sea evel estimate derived from the gridded sea level map. It can be applied at regional or local scale as a best estimate (better than no correction, since the rgional variation of the instrumental drift is unknown). See product manual for more details." ;
        tpa_correction:long_name = "TOPEX-A instrumental drift correction derived from altimetry and tide gauges global comparisons (WCRP Sea Level Bdget Group, 2018)" ;
        tpa_correction:scale_factor = 0.0001 ;
        tpa_correction:standard_name = "sea_surface_height_above_sea_level" ;
        tpa_correction:units = "m" ;
    int flag_ice(time, latitude, longitude) ;
        flag_ice:_FillValue = -2147483647 ;
        flag_ice:comment = "Ice Flag based on CDR OSI SAF products until 2016 (OSI-450), Interim products from 2016 (OSI-430-b) (Lavergne et al., 201). The flag corresponds to the 15% sea ice concentration." ;
        flag_ice:coordinates = "longitude latitude" ;
        flag_ice:flag_meanings = "data_on_sea data_on_ice" ;
        flag_ice:flag_values = 0, 1 ;
        flag_ice:grid_mapping = "crs" ;
        flag_ice:long_name = "Ice Flag for a 15% criterion of ice concentration" ;
        flag_ice:standard_name = "status_flag" ;

// global attributes:
        :Conventions = "CF-1.6" ;
        :Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
        :cdm_data_type = "Grid" ;
        :comment = "Sea Surface Height measured by Altimetry and derived variables" ;
        :contact = "http://climate.copernicus.eu/c3s-user-service-desk" ;
        :creator_email = "http://climate.copernicus.eu/c3s-user-service-desk" ;
        :creator_name = "Copernicus Climate Change Service (C3S)" ;
        :creator_url = "http://climate.copernicus.eu" ;
        :date_created = "2023-03-06T10:31:53Z" ;
        :date_issued = "2023-03-06T10:31:53Z" ;
        :date_modified = "2023-03-06T10:31:53Z" ;
        :geospatial_lat_max = 89.875 ;
        :geospatial_lat_min = -89.875 ;
        :geospatial_lat_resolution = 0.25 ;
        :geospatial_lat_units = "degrees_north" ;
        :geospatial_lon_max = 179.875 ;
        :geospatial_lon_min = -179.875 ;
        :geospatial_lon_resolution = 0.25 ;
        :geospatial_lon_units = "degrees_east" ;
        :geospatial_vertical_max = 0. ;
        :geospatial_vertical_min = 0. ;
        :geospatial_vertical_positive = "down" ;
        :geospatial_vertical_resolution = "point" ;
        :geospatial_vertical_units = "m" ;
        :history = "2023-03-06 10:31:54Z: Creation" ;
        :institution = "CLS, CNES" ;
        :keywords = "Oceans > Ocean Topography > Sea Surface Height" ;
        :keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
        :license = "http://climate.copernicus.eu/c3s-user-service-desk" ;
        :platform = "Sentinel-3A, Sentinel-6A" ;
        :processing_level = "L4" ;
        :product_version = "vDec2021" ;
        :project = "Copernicus Climate Change Service (C3S)" ;
        :references = "http://climate.copernicus.eu" ;
        :software_version = "7.1_DUACS_DT2021_baseline" ;
        :source = "Altimetry measurements" ;
```

:ssalto_duacs_comment = "The reference mission used for the altimeter inter-calibration processing is Topex/Poseidon between 1993-01-01 and 202-04-23, Jason-1 between 2002-04-24 and 2008-10-18, OSTM/Jason-2 between 2008-10-19 and 2016-06-25, Jason-3 between 2016-06-25 and 2022-02-09, and Sentinel-A from 2022-02-10." ;

:standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37" ;

:summary = "SSALTO/DUACS Delayed-Time Level-4 sea surface height and derived variables measured by multi-satellite altimetry observations ove Global Ocean." ;

:time_coverage_duration = "P1D" ;

:time_coverage_end = "2022-07-23T12:00:00Z" ;

:time_coverage_resolution = "P1D" ;

:time_coverage_start = "2022-07-22T12:00:00Z" ;

:title = "DT merged two satellites Global Ocean Gridded SSALTO/DUACS Sea Surface Height L4 product and derived variables" ;